Rakshasutra (SurakshaNet): A Zero-UI Cognitive-Assistive Wearable and Edge-Filtered BLE–LoRaWAN Architecture for Spatial Disorientation and Crowd Localization in Cellular-Denied Mountain Pilgrimages

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Abstract

Older adults affected by age-related memory decline, mild cognitive impairment (MCI), or early-stage dementia—alongside young children—experience acute spatial disorien- tation and wandering risk in high-density, unfamiliar environments such as mountain- ous religious pilgrimages. Existing smartphone-based assistive applications fail cogni- tively vulnerable individuals due to visual-motor interface complexity during panic, cel- lular network congestion, steep terrain shadowing, continuous metallic rain shelters, and mandatory mobile-phone restrictions inside inner temple sanctums. Simultaneously, pas- sive radio-frequency identification (RFID) checkpoints only log devotees within 1 to 3 me- ters of fixed turnstiles, leaving multi-kilometer transit corridors unmonitored. This pa- per presents the architectural design and analytical evaluation of Rakshasutra (Surak- shaNet), a cellular-independent, zero-user-interface (Zero-UI) cognitive-assistive wearable and reverse-tracking Internet of Things (IoT) network modeled across the 22.5 km multi- route Shri Mata Vaishno Devi (SMVD) mountain trek. Vulnerable individuals wear a reusable Bluetooth Low Energy (BLE) wristband featuring a single 3-second tactile SOS trigger and a psychophysically tuned acoustic-optical sensory beacon (3.2 kHz resonant piezo trans- ducer and 625 nm LED), while overhead shelter poles every 25 to 100 meters host split- antenna ESP32-S3 and SX1262 LoRaWAN relay nodes. To eliminate caregiver cognitive load and LoRaWAN Class A downlink latency, we propose a dual-layer spatial tether combining a direct peer-to-peer BLE caregiver-ward proximity tether with a LoRaWAN Class C 5-pole multicast connectionless wake protocol. To mitigate passive crowd stalking and packet replay, wearables broadcast a 15-byte AES-CMAC authenticated frame utilizing 15-minute rotating pseudonyms (Rolling_ID). To prevent payload overflow and pure-ALOHA chan- nel collapse in the Indian IN865 band during crowd surges of 3,000 concurrent users per cell, poles execute an in-RAM edge delta-filter combining data-rate-adaptive (DR-adaptive) arrival chunks with fixed 64-byte (512-bit) Bloom filter stationary heartbeats. We present explicit analytical derivations and audit reconciliations for BLE Link Layer PHY compatibil- ity, scan-window overlap bounds, Thevenin RC cold-cell transient voltage droop at $-5\circ$ C, and multi-channel IN865 offered load ($P_{loss} = 28.99\%$ under full 60-pole arrival/departure load, reducing to 21.35% with implicit departures and 12.4%–15.8% with optical-fiber hub offloading). All performance figures are explicitly distinguished as analytical estimates under stated component specifications to guide reproducible simulation and hardware validation. **Keywords:** Assistive Wearables, Cognitive Aging, Spatial Disorientation, Zero-UI Interaction, BLE 5.1, Bloom Filter, LoRaWAN IN865, AES-CMAC Pseudonymous Authentication, Edge Computing.

1 Introduction

Navigating crowded, acoustically noisy, and topographically complex public environments im- poses heavy demands on human working memory, spatial orientation, and executive atten- tion. In mass religious gatherings located in mountainous terrains—exemplified by the Shri Mata Vaishno Devi (SMVD) pilgrimage in Jammu and Kashmir, India, which attracts nearly 10 million devotees annually across 22.5 km of bifurcated mountain pathways—two cognitively and physically vulnerable demographics face severe safety risks: older adults (over 60 years) experiencing age-related cognitive decline, mild cognitive impairment (MCI), dementia-related wandering, or physical exhaustion, and young children (under 12 years). When an older adult with memory impairment or a young child becomes separated from their caregiver in a dense crowd, acute anxiety rapidly degrades their capacity to process complex instructions or operate graphical user interfaces (GUIs). Despite substantial infrastructural investments, existing localization and assistive technologies fail under four

operational and human-factors constraints unique to mountain shrines:

1. Cognitive Overload and Cellular Dependency of Smartphone Applications: Commer- cial crowd-sourced Bluetooth trackers (e.g., Apple AirTag, Reliance JioTag) and smartphone family-locator apps require active cellular backhaul, uninterrupted Global Navigation Satel- lite System (GNSS) sky visibility, uniform background app adoption, and multi-step screen interactions [14]. Along steep mountain gorges, cellular networks saturate or drop out en- tirely, and cognitively disoriented seniors cannot navigate smartphone menus or respond reliably to public-address loudspeaker announcements.

2. Inter-Checkpoint Blindness of Passive Turnstile RFID: While shrine authorities issue passive High-Frequency (HF) or Ultra-High-Frequency (UHF) RFID cards (Yatra Parchi) at base-camp registration counters, passive tags lack an internal power source and can only be energized within 1 to 3 meters of fixed portal readers at major gates. If a memory- impaired senior wanders onto a parallel mountain branch or collapses along a staircase between checkpoints, passive RFID provides zero localization or distress signaling capa- bility.

3. Electromagnetic Shielding by Metallic Shelter Infrastructure: To protect devotees from landslides and monsoon rain, over 70% of the mountain pathway is covered by grounded corrugated galvanized iron (GI) sheds. These steel roofs act as partial Faraday shields, introducing severe vertical radio-frequency (RF) attenuation that blocks GNSS satellite re- ception and degrades unoptimized links between ground-level wearables and hilltop re- ceivers.

4. Mandatory Mobile-Free Sanctum Queues: Prior to entering terminal sanctum queues and skywalk complexes (such as the main Bhawan complex), devotees are legally required to deposit all mobile phones and electronic bags inside cloakroom lockers. Consequently, even if security personnel recover a disoriented senior or child wearing a passive printed QR wristband, the caregiver's mobile phone is locked inside a cloakroom and cannot re- ceive voice calls or SMS alerts.

To address both the human-factors limitations of cognitively vulnerable users and the physical- layer constraints of metallic mountain corridors, we propose Rakshasutra (SurakshaNet), a cellular-independent, zero-user-interface (Zero-UI) cognitive-assistive wearable and edge- filtered BLE–LoRaWAN sensory network. Rather than burdening every individual with an ex- pensive, battery-intensive GPS-cellular terminal, the proposed architecture equips vulnera- ble wearers with a lightweight, reusable BLE beacon wristband (target bill of materials

INR 225–250 / $2.70–$3.00 USD) [4, 15] paired with a caregiver companion tag, while instru- menting overhead shelter poles every 25 to 100 meters with split-antenna BLE-to-LoRaWAN relay nodes [10, 11].

1.1 Capability Boundary and Evidentiary Scope

To maintain strict scientific transparency, two capability boundaries are stated explicitly through- out this manuscript:

- **Zone/Pole-Level Proximity Localization: Rakshasutra does not provide continuous centimeter- level or GPS coordinate tr**acking. Instead, the network resolves a wearer's last-seen smart pole and corridor zone (25–100 m spatial granularity) and bridges the final 10–30 m recovery gap using localized acoustic-optical sensory cues on the wearable.
- **Separation of Analytical Models from Physical Measurements: All numerical evalu- ations presented in this paper repres**ent analytical estimates derived from stated mathe- matical models and official component specifications (Nordic Semiconductor, Semtech, LoRa Alliance, and Panasonic Energy datasheets). Where baseline design targets conflict with simplified physical models—such as BLE Legacy vs. Coded PHY packet formatting, asyn- chronous scan overlap probabilities, coin-cell RC pulse transients at $-5\circ$ C, and multi-channel pure-ALOHA offered loads—we present the complete mathematical reconciliation and iden- tify the exact bench and field validation protocols required prior to deployment.

1.2 Summary of Architectural and Analytical Contributions

1. Zero-UI Multi-Modal Sensory Interface and Dual-Layer Spatial Tether: We design a screen-free wearable interface combining a 3-second tactile SOS trigger, a presbycusis- tuned 3.2 kHz resonant piezo and 625 nm LED recovery beacon, a Layer-1 Direct Peer-to- Peer BLE Caregiver Tether, and a Layer-2 LoRaWAN Class C 5-Pole Multicast Wake Protocol. We derive the exact scan-window overlap conditions required to eliminate the 97.5% miss rate of naive sparse scanning.

2. Zero-Trust Pseudonymous BLE Frame with Anti-Linkability Protection: We formulate a compact 15-byte non-connectable BLE payload integrating a 32-bit rotating pseudonym (Rolling_ID updated every 15 minutes), an ephemeral salt/nonce to prevent cleartext epoch-counter linkability, and a 32-bit truncated AES-CMAC Message Integrity Code (MIC) [3] supporting O(1) server-side lookup in Redis.

3. DR-Adaptive Edge Delta-Filtering and 64-Byte Bloom Filter Heartbeats: We derive the Semtech SF7 Time-on-Air and multi-channel pure-ALOHA collision models for the Indian IN865 band [1, 2, 9] and design an ESP32-S3 edge state machine that suppresses duplicate BLE pings in RAM, adapts arrival batch sizes to the active LoRaWAN Data Rate (DR0–DR5), and compresses 60-second stationary presence into a fixed 64-byte (512-bit) Bloom filter [6, 7] with a restricted set-intersection proof that prevents false spatial jumps.

4. Component-Resolved Power and Cold-Cell Thevenin RC Transient Analysis: We recon- cile BLE Link Layer airtime across LE 1M Legacy Advertising (nRF52811

at +4 dBm) and LE Coded PHY Extended Advertising (nRF52833 at +8 dBm), and solve the exact Thevenin RC differential equation (= 4.20 ms) for a CR2450 coin cell at −5 ◦ C (Rint = 42 Ω) [16].

2 Related Work and Comparative State-of-the-Art

Prior approaches to wandering prevention and crowd localization span five technological paradigms, compared in Table 1:

1. Dementia Wandering GNSS/Cellular Wearables: Commercial GPS watches rely on satel- lite visibility and cellular modems to enforce geofences. Beneath corrugated steel moun- tain shelters, GNSS receivers lose lock, cellular towers saturate during festival surges, hardware costs exceed INR 2,500 ($30 USD), and batteries deplete within 18 to 30 hours.

2. Printed QR Bands and Passive Turnstile RFID: Deployed at flat-terrain festivals and base- camp entry gates, passive bands are strictly reactive: a guard must physically spot a dis- tressed individual and scan their wrist. When caregivers lock their mobile phones inside temple cloakrooms, calling a phone number printed on a QR band is impossible.

3. Industrial BLE-over-LoRaWAN Trackers: Commercial industrial badges (e.g., Abeeway, SenseCAP) place both the BLE scanner and the LoRaWAN transceiver inside the wearable while mounting static beacons on walls. This inverted architecture increases wearable cost to INR 3,000–4,500 ($35–$55 USD) and adds bulk and weight, preventing mass issuance.

4. Naive BLE-to-LoRaWAN Forwarding: Academic prototypes that forward raw BLE MAC addresses over LoRaWAN Class A nodes [10, 12] suffer from severe pure-ALOHA packet collisions when crowd density exceeds 1,000 tags per cell, expose static MAC addresses to unauthorized tracking, and incur up to 68 seconds of Class A + GATT downlink delay.

3 Topographical Graph, RF Channel, and Psychophysical Sensory Model

1.3 Bifurcated Mountain Graph Formulation and Pole Arithmetic Reconcilia- tion

We model the 22.5 km SMVD pilgrimage network as a directed, weighted, bifurcated graph G = (V, E, W):

• **V = {p1 , p2 , . . . , pN } denotes the set of Smart Pole Relay Nodes positioned at topological coordinates spanning a**ltitudes h(pk) [762 m, 2017 m] (2,500 ft to 6,619 ft).

• **E V × V represents walkable and electric-vehicle pathway edges across six operational zones: Zone 1 (Old Track & sta**ircases, 6.0 km), Zone 2 (Tarakote Marg parallel corridor, 7.0 km), Zone 3 (Adhkuwari Junction queue hub, 0.5 km), Zone 4 (Himkoti EV corridor & Sanjichhat ridge, 6.5 km), Zone 5 (Bhawan sanctum, Skywalk & Cloakrooms, 1.0 km), and Zone 6 (Bhairon Ghati climb, 1.5 km).

• **W : E → R+ assigns the physical path length w(pa , pb) in meters to each edge.**

Audit Reconciliation of Route Pole Arithmetic (Claim C01): In earlier architectural drafts, Zone 4 (6.5 km) was listed with 100 m nominal spacing alongside a count of 25 poles (sum- ming to N = 255 poles), which only covers 2.5 km of shelter/curve segments. To ensure strict arithmetic consistency across the 6.5 km Zone 4 corridor at 100 m nominal spacing (65 intervals), Table 2 reports both the 255-pole partial-corridor baseline (where only shelter hubs and sharp curves in Zone 4 are instrumented) and the 295-pole full-corridor specification (where all 6.5 km of Zone 4 are instrumented every 100 m). In both configu- rations, the 60 highest-density hub poles in Zones 3 and 5 utilize wired W5500 Optical Fiber LAN backhaul, leaving 195 (partial) to 235 (full) poles on LoRaWAN IN865 across 6 gateways (32.5 to 39.2 LoRa poles/gateway cell). Because the pathway bifurcates at Banganga (p1 → Zone 1 vs. Zone 2) and Adhkuwari (Zone 3 → Himkoti vs. Sanjichhat), simple subtraction of pole indices |a − b| produces invalid separation alerts. The geodesic track distance between a vulnerable wearer at pole pw and a caregiver at pole pc is computed over all valid path sequences Π(pw , pc) in G: X dG (pw , pc) = min w(e) (1) Π(pw ,pc) e

1.4 Zero-UI Psychophysical and Bio-Acoustic Sensory Interface Design

To support cognitively overloaded seniors and young children without visual-motor screen interaction, the wearable utilizes three low-cognitive-load sensory channels:

1. Single-Action Tactile SOS Trigger: A recessed tactile push-button requiring a continuous 3-second hold (Thold = 3.0 s) prevents accidental activation from crowd contact while re- maining accessible during medical distress or disorientation. Upon trigger, the wearable shifts its BLE advertising interval from 800 ms to a 200 ms high-priority burst.

2. Caregiver Peer-to-Peer Spatial Tether Alert: The Caregiver Companion Tag (a lanyard/key- chain receiver carried by the guardian) monitors the wearer's BLE transmissions directly and triggers an onboard acoustic-visual alert when the wearer drifts beyond the local link margin for 4 consecutive scan windows (19.2–20.0 s).

3. Presbycusis-Tuned Acoustic and Photometric Recovery Beacon: Designing an acous- tic recovery cue for older adults inside a noisy crowd (Lamb 72–78 dBA) requires bal- ancing two opposing bio-acoustic constraints: human vocal chatter concentrates spectral masking energy between 500 Hz and 1.5 kHz, whereas age-related sensorineural hearing loss (presbycusis) severely attenuates auditory sensitivity above 4.5–6.0 kHz. Rakshasutra drives a resonant piezo transducer tuned to f0 = 3.2 kHz—coinciding with the primary acoustic resonance of the human external auditory meatus (2.7–3.4 kHz) while remaining below the steep presbycusis roll-off. Modulating the 3.2 kHz carrier with a 2.0 Hz temporal envelope (100 ms ON / 400 ms OFF, 20% duty cycle) exploits human temporal dip-listening and prevents neural adaptation, while a 625 nm high-brightness red LED

Table 1: Comparison of Assistive Localization Paradigms for Vulnerable Users in Dense Mountain Corridors

Attribute	Passive RFID / QR	GPS + Cellular	Consumer BLE Tag	Naive BLE–LoRa	Rakshasutra
Wearer UI demand	Passive; none	Screen/app; charging	Smartphone/app dependent	Passive beacon	Zero-UI; 3 s SOS + sensory cue
Inter-gate tracking	Gate-only, approx. 1–3 m	Open-sky GNSS; weak under sheds	Depends on nearby phones	Low-density only	Pole/zone-level, 25–100 m
Caregiver alert	None	30–60 s; live 4G needed	No separation tether	Class A downlink delay	Direct BLE tether + Class C wake
Phone-banned zones	No caregiver alert path	Phone unavailable in sanctum	Phone may be locked away	No caregiver interface	Caregiver tag + RFID kiosks
Shelter bypass	Portal gates only	GNSS/LTE obstruction	Multipath limitations	Single-antenna limitations	Split-antenna architecture
Anti-stalking security	Static identifier	SIM/network security	Rotating identifier varies	Cleartext MAC risk	15-min Rolling_ID + CMAC
3,000-tag surge	Gate-only coverage	Base-station saturation risk	Not applicable	High ALOHA loss risk	DR-adaptive batches + Bloom filter
Target BOM / runtime	INR 10; single-use	INR 2,500+; 18–30 h	INR 1,000–2,900; approx. 1 yr	INR 350; 3–6 mo	INR 225–250; estimated 10–20+ mo

Table 2: SMVD 22.5 km Multi-Route Reference Topology and Reconciled Pole Allocation

Zone	Corridor segment	Length	Altitude	Partial	Full	Backhaul / edge rule
1	Banganga–Adhkuwari (Old Track)	6.0 km	2,800–4,800 ft	85	85	LoRa Class C; split antenna; $H_{\text{req}} = 1$
2	Tarakote Marg (parallel route)	7.0 km	2,800–4,800 ft	70	70	LoRa Class C; solar + $LiFePO_4$; $H_{\text{req}} = 1$
3	Adhkuwari junction hub	0.5 km	4,800 ft	20	20	Fiber LAN (W5500) + LoRa failover; $H_{\text{req}} = 3$
4	Himkoti EV / Sanjichhat ridge	6.5 km	4,800–6,200 ft	25 (hubs)	65	LoRa Class C; EV transit; $H_{\text{req}} = 1$
5	Bhawan sanctum / cloakrooms	1.0 km	5,200 ft	40	40	Fiber LAN (W5500); 6 RFID kiosks; $H_{\text{req}} = 3$
6	Bhairon Ghati steep climb	1.5 km	5,200–6,619 ft	15	15	LoRa Class C; cold specification; $H_{\text{req}} = 1$
Total reference graph		**22.5 km**	**2,500–6,619 ft**	**255**	**295**	**60 fiber hub poles + 195/235 LoRa poles**

(120 ◦ viewing an- gle) provides a high-contrast visual cue under dimly lit rain shelters.

1.5 Split-Antenna Faraday Shelter Bypass and Analytical Link Budget

Corrugated galvanized iron (GI) rain shelters (hshed 3.2 m, width wshed 4.5 m) obstruct vertical RF propagation. In our analytical link-budget model, we parameterize roof penetra- tion loss as LGI (2.44 GHz) = 24.1 dB and LGI (866 MHz) = 18.4 dB (Claim C13; note that these represent literature/model parameters rather than audited site measurements). Mounting a relay node entirely above the steel roof blinds the 2.4 GHz BLE receiver to wearables below, whereas mounting it entirely underneath with an internal antenna attenuates the 866 MHz LoRa uplink to hilltop gateways. Rakshasutra bypasses LGI on both links via a Split-Antenna Pole Architecture:

• **Under-Shed 2.4 GHz BLE Sub-Link: The weatherproof IP65 ESP32-S3 enclosure is mounted underneath the steel roof truss a**t 3.0 m height, maintaining unobstructed downward line- of-sight to wearables. Received power Prx,BLE (d) at distance d is modeled via the log- distance path-loss equation:

d Prx,BLE (d) = Ptx,BLE + Gtx + Grx − P L(d0) − 10nshed log10 − Lbody − X (2) d0

where P L(d0 = 1 m) = 40.2 dB, nshed = 2.15 (modeling waveguide reflections under the steel sheet), Lbody = 14.0 dB, and X N (0, 2) with = 4.2 dB.

• **Over-Shed 865–867 MHz LoRa Uplink: A 1 m RG-316 coaxial pigtail (Lcoax = 0.9 dB) routes the SX1262 RF output through a** weatherproof gland to a 3 dBi vertical whip an- tenna mounted above the steel roof ridge. With Ptx,LoRa = +14 dBm, gateway antenna gain Ggw = 5.8 dBi, and SX1302 sensitivity Srx,SF7 = −126.5 dBm (−131.5 dBm at SF9) [17], the analytical maximum allowable path loss is MAPL = 145.4 dB.

Audit Reconciliation of BLE PHY Mode and SoC Transmit Power (Claims C02 & C03): A critical protocol audit finding is that the Bluetooth Core Specification restricts legacy adver- tising PDUs (ADV_NONCONN_IND, 31-byte maximum payload) to the 1 Mbps LE 1M PHY; LE Coded PHY (S = 2 or S = 8) is only permitted inside Extended Advertising PDUs (ADV_EXT_IND + AUX_ADV_IND) [4, 5]. Furthermore, Nordic Semiconductor's product specification for the nRF52811 caps maximum transmit power at +4 dBm (7.0 mA DC/DC TX current, −104 dBm sensitivity at 125 kbps Coded PHY) [15], whereas +8 dBm output requires the pin-compatible nRF52833 SoC (14.2 mA DC/DC TX current). Table 3 formalizes both valid hardware/protocol configurations.

4 Zero-Trust Cryptographic Protocol and Synchronized Wake Ar- chitecture

Table 3: Reconciled Wearable BLE Hardware and Link-Layer PHY Configurations

Parameter	Config A: Standard hub/track	Config B: Long-range extended
SoC / maximum TX power	Nordic nRF52811, +4 dBm	Nordic nRF52833, +8 dBm
BLE link-layer PDU	Legacy ADV_NONCONN_IND (3 channels)	Extended ADV_EXT_IND + AUX_ADV_IND
PHY	LE 1M, 1 Mbps	LE Coded, S=8, 125 kbps
RF bit airtime	264 μs/channel $\times 3 = 0.792$ ms	$3 \times 0.976 + 2.384 = 5.312$ ms
Estimated event duration	1.15–1.25 ms, including ramp-up / IFS	5.80–6.20 ms, including ramp-up / IFS
TX current	7.0 mA at +4 dBm, 3.0 V	14.2 mA at +8 dBm, 3.0 V
Analytical range target	28–35 m (−96 dBm RX sensitivity)	80–92 m (−103 dBm coded sensitivity)

1.6 Adversarial Threat Model and Key Hierarchy (Claim C12)

Tracking children and memory-impaired older adults requires defense against four adversar- ial threats:

• **Adversary A1 (Passive Crowd Stalker): Deploys BLE sniffers along the trek to link advertisements across poles and tr**ack a specific individual or infer family groupings. • Adversary A2 (Packet Replay / SOS Flooder): Captures valid BLE advertisements or SOS frames and replays them at later times or different poles to trigger false alarms. • Adversary A3 (Unauthorized Downlink Trigger): Broadcasts forged BLE wake packets to activate a wearer's acoustic-optical beacon or drain their coin-cell battery. • Adversary A4 (Physical Node Tampering): Captures a smart pole or lost wristband to extract cryptographic keys via debug readback.

To mitigate A4 , the on-premise server stores a 128-bit master key Kmaster inside an encrypted keystore. During provisioning, each wearable u receives a unique derived 128-bit key [3]:

Ku = AES-CMACKmaster UIDu k Saltsys (3) and MCU debug readback protection (APPROTECT) is permanently enabled [15]. Smart Poles never store Kmaster or any Ku ; poles act strictly as untrusted relays using ESP32-S3 Secure Boot v2, Flash Encryption, and enclosure tamper-switch RAM zeroization. Production deployments must never use public RFC test vectors as operational keys.

1.7 15-Byte Signed BLE Frame and Anti-Linkability Reconciliation (Claims C12 & C17)

To eliminate slow two-way BLE clock-synchronization writes at registration counters, each wearable maintains an internal 16-bit monotonic epoch counter ci [0, 65535] (Slot_Counter) clocked by its 32.768 kHz crystal, incrementing every Tslot = 900 s (15 minutes). At each epoch boundary, the wearable updates both its 48-bit BLE Random Static MAC address Mu,i and its 32-bit pseudonym (Rolling_ID): Rolling_IDu,i = Trunc32 (AES-CMACKu (0x01 k ci)) (4) Mu,i = Trunc46 (AES-CMACKu (0x02 k ci)) k 0b11 (5)

Audit Fix for Epoch-Counter Linkability (Claim C12): In the baseline v5.0 specification, transmitting the raw per-device counter ci in cleartext ('Bytes 3–4') allowed a passive observer (A1) to correlate ci → ci + 1 across a 15-minute rotation boundary if two wearables had distinct counter offsets. In the reconciled v5.1 protocol, because the server already precomputes expected Rolling_IDs across the 3-slot sliding window Wu (t) = {ĉu (t) − 1, ĉu (t), ĉu (t) + 1} in Redis, the wearable does not transmit cleartext ci during the trek. Instead, 'Bytes 3–4' carry an **encrypted/masked epoch verifier or 16-bit ephemeral sub-slot nonce rk ** (rotated every 30 s): Pu,i = 0xFF5A | {z } k V type k rk k Rolling_IDu,i k Fstat k Bmv k MICu,i (6) | {z } |{z} | {z } |{z} |{z} | {z } Comp_ID (2B) 1B Nonce (2B) 1B 1B 4B 4B where MICu,i is a 32-bit truncated AES-CMAC computed over the header, internal epoch ci , ephemeral nonce rk , status flags Fstat , and battery voltage Bmv :

MICu,i = Trunc32 AES-CMACKu Vtype k ci k rk k Rolling_IDu,i k Fstat k Bmv (7)

At counter registration, placing the band on an NFC/BLE pairing pad wakes the magnetic reed switch and captures its initial broadcast frame in < 1 s, storing (cinit , tpair) in Redis. At server timestamp t, the expected epoch counter is: t − tpair

ĉu (t) = cinit + mod 65536 (8) 900 To tolerate ±20 ppm crystal drift (±1.73 s/day), the server precomputes Rolling_IDu,k for Wu (t) = {ĉu (t) − 1, ĉu (t), ĉu (t) + 1} in an in-memory Redis hash table, resolving identities in O(1) time and rejecting replayed (rk , MICu,i) tuples.

1.8 Scan-Window Overlap Audit and Synchronized Wake Protocol (Claims C04 & C05)

Audit Identification of Sparse Asynchronous Scan Deficit: In the baseline v5.0 proposal, the wearer advertised every Tadv = 800 ms, while the Caregiver Tag scanned for Tscan = 5 ms every 5 s (alarming after 4 misses = 20 s) and the Wearer Band scanned for 5 ms every 4 s dur- ing Find Mode (poles broadcasting every 50 ms). As established in the Day 1 audit, if two peri- odic processes are unsynchronized (random phase), a single 5 ms receive window overlaps an 5 ms 800 ms periodic transmission with idealized single-channel probability p1 800 ms = 0.00625. Across 4 windows, the cumulative probability of catching at least one beacon is only:

5 4

Poverlap, async (4) = 1 − 1 − 2.48% (9) 800 which would cause continuous false separation alarms! Similarly, a 5 ms wearable scan win- dow (0.125% duty cycle)

Table 4: Reconciled 15-Byte Authenticated BLE Advertisement Frame

Bytes	Field	Size	Specification
0–1	Company ID	2 B	0xFF5A test ID; replace with assigned Bluetooth SIG ID in production.
2	V_{type}	1 B	Bits 0–3: protocol version (0x5); bit 4: role (0=wearer, 1=caregiver).
3–4	r_k (sub-nonce)	2 B	Ephemeral nonce rotated every 30 s; replaces cleartext epoch counter.
5–8	Rolling_ID	4 B	32-bit pseudonym derived from AES-CMAC; rotated every 15 min.
9	F_{stat}	1 B	Bit 0 SOS; bit 1 tether lost; bit 2 beacon active; bit 3 low battery.
10	B_{mv}	1 B	Supply voltage in 20 mV steps; $V_{\text{batt}} = B_{mv} \times 0.02$ V.
11–14	MIC	4 B	32-bit truncated AES-CMAC over authenticated frame fields.

can fall entirely inside the 45 ms silent gap between 50 ms pole wake broadcasts.

Reconciled v5.1 Timing Architecture: We resolve both timing deficits analytically:

1. Crystal-Synchronized Caregiver Tether (Claim C04): Because the Caregiver Tag and Wearer Band are paired together on the counter pad (where the caregiver tag receives the wearer's next 96 pre-computed Rolling_IDs covering 24 hours = 384 bytes), the Caregiver Tag synchronizes to the wearer's initial broadcast phase t0 . Over a re-sync interval of Tsync = 4.8 s (6 × 800 ms), worst-case relative clock drift between two ±20 ppm crystals is ±40 ppm × 4.8 s = ±0.192 ms. Accounting for the Bluetooth Link Layer pseudo-random advDelay [0, 10] ms [5] and radio startup, the Caregiver Tag opens a phase-locked window of Tscan,cg = 14.0 ms every 4.8 s (re-centering phase upon each received packet, or expanding to a continuous scan on the rechargeable/larger-cell caregiver lanyard unit if 2 consecu- tive windows miss). Four missed synchronized windows trigger the local caregiver alert at 4 × 4.8 s = 19.2 s.

2. Deterministic-Overlap Connectionless Find Mode (Claim C05): All Smart Poles operate in continuous-receive LoRaWAN Class C mode (SX1262 RX draws 4.6 mA [17] alongside the 75 mA ESP32-S3 BLE scanner). When a caregiver or guard triggers Find Mode for a wearer last seen at pole pk , the server computes an 8-byte wake frame containing a 16- bit monotonic command sequence counter Seqcmd (preventing downlink replay A3) and a 24-bit token:

Tcmd = Trunc24 AES-CMACKu 0xFA k Rolling_IDu,i k Seqcmd k Cmd (10)

and multicasts the command simultaneously to the 5-pole topological neighborhood N (pk) = {pk−2 , pk−1 , pk , pk+1 , pk+2 }. Each of the 5 poles broadcasts this wake frame as a non-connectable BLE advertisement at the minimum spec interval Twake_adv = 20.0 ms for 15 s. By setting the wearable's passive scan window to Tscan,w = 22.5 ms every Tscan_int = 4.0 s, we enforce the deterministic overlap condition Tscan,w > Twake_adv + Tpkt , guaranteeing geometric in- tersection within the very first scan cycle (mean analytical wake delay 2.5 s, single-cycle bound 4.5 s under zero channel loss).

5 Edge Delta-Filtering, LoRaWAN IN865 Capacity, and Kinematic Models

1.9 Exact IN865 Time-on-Air Derivation (Claim C09)

In the Indian IN865–867 MHz band [1, 9], LoRaWAN gateways operate with bandwidth BW = 125 kHz, coding rate CR = 1 (4/5), explicit header IH = 0, CRC enabled (CRC = 1), low-data-rate optimization DE = 0 (for SF7–SF10), and preamble length npre = 8 symbols. By Semtech's 2SF physical-layer specification [2], the symbol duration is Tsym = BW (1.024 ms at SF7). For a total PHY payload of P L = 13 + Bapp bytes (including the 13-byte LoRaWAN MAC header and MIC), the general Time-on-Air equation is:

8 · P L − 4 · SF + 28 + 16 · CRC − 20 · IH Tair (Bapp) = Tsym npre + 4.25 + 8 + max (CR + 4), 0 4(SF − 2 · DE) (11) Substituting SF = 7, CRC = 1, IH = 0, DE = 0, CR = 1, npre = 8, the numerator inside the ceiling operator simplifies algebraically to 8 · P L−28+28+16 = 8(13+Bapp)+16 and the denominator to 4(7) = 28: 8(13 + Bapp) + 16 Tair,SF7 (Bapp) = 1.024 ms × 20.25 + 5 (12) Evaluating Equation (12) yields the exact SF7 packet airtimes used in our traffic ledger:

- **Priority SOS Frame (Bapp = 15 B = P L = 28 B): 20.25 + 5d240/28e = 65.25 symbols = 66.82 ms.**
- **64-Byte Bloom Filter Heartbeat (Bapp = 64 B = P L = 77 B): 20.25 + 5d632/28e = 135.25 symbols = 138.50 ms.**
- **Mean 13.6-Tag Arrival Batch (Bapp = 112 B = P L = 125 B): 20.25 + 5d1016/28e = 205.25 symbols = 210.18 m**s.
- **Max 25-Tag Arrival Chunk (Bapp = 200 B = P L = 213 B): 20.25 + 5d1720/28e = 330.25 symbols = 338.18 m**s.

1.10 DR-Adaptive Edge Delta-Filtering and 64-Byte Bloom Filter Engine (Claims C10 & C11)

Audit Fix for IN865 Payload Limits Across Data Rates (Claim C10): Under LoRa Alliance Regional Parameters RP002-1.0.5 for IN865 [1], the maximum application payload (assuming empty FOpts) is 222 bytes at DR4/DR5 (SF8/SF7), 115 bytes at DR3 (SF9), and 51 bytes at DR0–DR2 (SF12–SF10). Consequently, a fixed 200-byte (25-record) arrival batch is valid only at DR4/DR5 and will be rejected by the MAC layer if Adaptive Data Rate (ADR) steps a pole down to DR3 or lower. Furthermore, if MAC commands are

piggybacked in FOpts (LFOpts [1, 15] bytes), maximum payload shrinks accordingly. In our reconciled edge state machine (ActiveTagTable[1024] in ESP32-S3 SRAM), each pole dynamically computes its maximum record chunk size Cmax (DR, LFOpts) every 15 seconds:

Lmax,MAC (DR) − LFOpts − 4

Cmax (DR, LFOpts) = min 25, (13)

yielding Cmax = 25 tags (200 B) at DR4/DR5, Cmax = 12 tags (96 B) at DR3, and Cmax = 5 tags (40 B) at DR0–DR2. 64-Byte Bloom Filter Heartbeat and Zero-False-Jump Proof (Claim C11): To detect if an older adult has collapsed or remained motionless at a single pole for > 20 minutes (station- ary dwell alert), poles report currently present tags every 60 s. Sending a raw list of 4-byte Rolling_IDs overflows the 222 B DR5 ceiling whenever n > 55 devotees shelter under a pole during rain (and overflows DR3 at n > 28). Instead, each pole hashes all n active Rolling_IDs into a fixed m = 512-bit (64-byte) Bloom filter Bk using k = 3 double-hashing functions [6, 7]:

hj (x) = (CRC16(x) + j · FNV1a32 (x)) mod 512, j {0, 1, 2} (14)

For m = 512, k = 3, the standard false-positive probability approximation is: k 2.60% at n = 60 stationary tags/pole −kn/m Pfp (n) 1 − e = 8.70% at n = 100 stationary tags/pole (15) 32.93% at n = 200 stationary tags/pole

As noted in our Day 1 audit (Claim C11), a Bloom filter is an aggregate membership hint, not standalone proof of individual location. Let Sauth (pk , t) denote the set of tags already authenticated at pole pk in Redis via an exact, cryptographically verified arrival tuple (FPort 1). Because the server queries Bk strictly over the restricted intersection: 2 ^ h i Sdwell (pk , t) = u Sauth (pk , t) Bk hj (Rolling_IDu,i) = 1 (16) j=0

a false positive in Bk can slightly delay the 90-second departure timeout of a tag u Sauth (pk , t) that just walked away (until u is authenticated at adjacent pole pk±1), but **can never teleport an unauthenticated tag v / Sauth (pk , t) to pole pk **. If n > 120 at a single pole, the pole sets a header bit and splits the filter across two 64-byte epochs.

1.11 Reconciled LoRaWAN IN865 Offered-Load and Collision Model (Claim C09)

We now present the fully reconciled Day 1 mathematical audit of single-cell LoRaWAN IN865 pure-ALOHA offered load G and packet collision loss Ploss = 1 − e−2G [12] for a surge of Ncell = 3, 000 concurrent wearables moving across Kpoles = 60 poles with mean inter-pole pass time pass = 55 s across M = 8 gateway channels at SF7:

1. 60-Second Bloom Heartbeat Load (EHB): Each of the 60 poles transmits one 64-byte Bloom filter (Tair = 0.1385 s) every 60 s: 0.1385 s EHB = 60 × = 0.1385 Erlangs (17) 60 s

2. 15-Second Arrival Batch Load (Earr): With Ncell = 3, 000 tags and pass = 55 s, aggregate cell arrival rate is cell = 3000 55 s = 54.55 tag arrivals/s. Spread evenly across 60 poles over a 15 s flush window, each pole accumulates an average of $\bar{c}$ = 54.55×1560 = 13.64 arrivals/batch (109.1 B payload; Tair 0.205–0.210 s at SF7): 0.205 s Earr = 60 × = 0.8200 Erlangs (18) 15 s

3. 15-Second Explicit Departure Batch Load (Edep): If poles also transmit explicit 4-byte departure lists (FPort 2, 13.64 × 4 B 55 B = Tair 0.1025 s every 15 s): 0.1025 s Edep = 60 × = 0.4100 Erlangs (19) 15 s

Table 5 reports the exact, reconciled offered load G = EM total and pure-ALOHA collision proba- bility Ploss = 1 − e−2G across all baselines and architectural variants (Claim C09): Key Audit Takeaway on Table 5: In earlier drafts, row 4 reported G = 8.6% (Ploss = 15.8%) for a 60-pole cell, which understated the 60-pole arrival load (0.55 E vs. 0.820 E). As shown in Rows 4a–4c of Table 5, when all 60 poles transmit over LoRa with explicit departures, true pure-ALOHA loss at 3,000 tags is 28.99% (G = 17.11%). Replacing explicit departure frames with server-side implicit departures (when tag u is authenticated at pole pk+1 , the server au- tomatically closes its dwell at pole pk , saving 33.3% of event traffic and 30.0% of total traffic) reduces 60-pole loss to 21.35% (G = 11.98%). Finally, because the 60 highest-density poles in Zones 3 and 5 offload traffic directly to wired W5500 Optical Fiber LAN (Table 2), each of the 6 LoRaWAN gateways serves only 32.5 to 39.2 LoRa poles, bringing per-cell LoRa loss down to 14.77% (G = 7.99% at 40 poles/cell). Note that pure ALOHA (1 − e−2G) remains a simplified baseline that must be paired with a capture-aware and inter-SF interference simulation [13].

1.12 Constant-Velocity (a = 0) EV Transit Opportunity Model (Claim C14)

Along Zone 4 (Himkoti Marg), older adults frequently travel in electric Battery Cars cruising at a steady velocity vev = 20 km/h = 5.56 m/s. Because linear acceleration during constant- speed cruising is zero (a = dv/dt = 0), motion-gated wearables that rely on an accelerometer

interrupt to switch into fast-ping mode remain in slow stationary mode (2–5 s) and cross a pole's 25 m core detection zone in: 25 m Tcore = = 4.50 s (20) 5.56 m/s failing multi-hit filters (hitCount 3). Rakshasutra removes the accelerometer, locks the active advertising cadence to Tadv = 800 ms, and sets Hreq = 1 on transit corridors (relying on the 32-bit AES-CMAC MIC to reject random RF noise). Audit Qualification of Transit Capture Probability (Claim C14): A 4.50 s transit window pro- vides Nopp = b4.50/0.80c = 5 to 6 nominal advertisement opportunities. In earlier drafts, this was reported as a single 99.4% capture figure (which corresponds to an idealized independent binomial model Pdet = 1 − p5miss = 99.4% at per-frame miss probability pmiss = 0.35). However, in real vehicular tran-

Table 5: Reconciled Single-Cell LoRaWAN IN865 Offered Load and Pure-ALOHA Loss

Forwarding architecture	500 tags	1,000	2,000	3,000	Audit / payload note
Raw BLE forwarding (2 s ping)	188% / 97.7%	376% / 99.9%	752% / 100%	1128% / 100%	Channel collapse
Unbatched arrival + raw heartbeat	6.5% / 12.2%	13.0% / 22.9%	26.0% / 40.5%	39.0% / 54.2%	Exceeds 222 B limit above 55 tags/pole
15 s batch + raw-ID heartbeat	4.8% / 9.1%	8.1% / 15.0%	14.8% / 25.6%	21.5% / 34.9%	Exceeds 222 B limit above 55 tags/pole
Rakshasutra, 60 poles, explicit departures	4.1% / 7.9%	6.9% / 12.9%	12.0% / 21.3%	17.11% / 28.99%	1.3685 E total offered load
Rakshasutra, 60 poles, implicit departures	3.1% / 6.0%	5.1% / 9.7%	8.5% / 15.6%	11.98% / 21.35%	Omits 0.41 E departure load
Rakshasutra, 40 LoRa poles + fiber hubs	2.1% / 4.1%	3.4% / 6.6%	5.7% / 10.8%	7.99% / 14.77%	60 hubs on W5500 fiber LAN

Each cell reports offered load G / pure-ALOHA loss P_{loss}.

sit, packet losses are correlated due to body orientation, metallic vehicle frame blockage, and co-channel interference. Under a correlated Gilbert-Elliott burst-loss or shadowing model where effective independent trials reduce to Neff [2, 5], analytical cap- ture probability spans Pdet [87.5%, 99.4%], requiring empirical vehicular field trials across passenger seating positions.

6 Wearable Power Budget and Sub-Zero (−5 ◦ C) Thevenin RC Tran- sient Audit

1.13 Component-Resolved Wearable Current Model (Claims C06 & C07)

Table 6 compares the baseline analytical current sum (29.74 µA) against the component-resolved v5.1 estimates incorporating our reconciled BLE PHY modes (Table 3) and the deterministic 22.5 ms Find-Mode scan window (Claim C05): Ttx Itx Tscan,w Irx Iavg = Isleep + + + Icrypto (21) Tadv Tscan_int

Reconciliation of CR2450 Autonomy Ratios (Claims C07 & C15): Panasonic's CR2450 specifi- cation lists a nominal capacity of Qnom = 620 mAh under a 0.2 mA continuous drain at 20 ◦ C [16]. As documented in our Day 1 audit: mAh • At the baseline 29.74 µA current, the pure capacity-to-current ratio is 0.02974 620 mA×730.5 h/mo = 28.5 months ideal (20.0 months with a 30% cold derating to 434 mAh; the baseline paper's 11.1 months figure corresponded to a conservative 241 mAh net usable capacity budget after deducting alarm activations and cutoff margin). • Under the reconciled Config A (37.74 µA) with deterministic 22.5 ms scanning, the ideal continuous runtime ratio is 22.5 months at 620 mAh, 15.7 months at 434 mAh, and 8.7 months continuous (13.8 months under a 14 h/day active trek + 10 h/day 0.60 µA rack sleep cycle) at a conservative 241 mAh usable cold budget. • Under Config B (130.63 µA, Coded PHY S = 8 at +8 dBm), continuous runtime is 6.5 months at 620 mAh (4.5 months at 434 mAh), demonstrating the exact energy-vs.-range trade-off between 'LE 1M' and 'LE Coded S=8'.

1.14 Thevenin RC Cold-Cell Transient Audit at −5 ◦ C (Claim C08)

At −5 ◦ C winter conditions at Bhairon Ghati (6,619 ft), the internal resistance of a Li-MnO2 CR2450 coin cell increases substantially (modeled as Rint = 42 Ω with open-circuit voltage Voc =

1.15 V). In earlier drafts, a C = 100 µF tantalum hold-up capacitor across VDD was claimed to restrict voltage droop to V = 0.27 V (Vterm = 2.68 V) during a 100 ms ON / 400 ms OFF (20% duty cycle) Iload = 32 mA piezo/LED pulse. Audit Refutation and Exact Differential Equation Solution (Claim C08): For the simplified Thevenin source (Voc = 2.95 V, Rint = 42 Ω) buffered by C = 100 µF, the RC time constant is:

= Rint · C = 42 Ω × 100 µF = 4.20 ms (22)

In the ideal capacitor-only limit, a 32 mA current over 100 ms would droop by V = I · t C = 32 V, proving that a 100 µF capacitor cannot sustain a 100 ms pulse (23.8). Instead, during any pulse of duration ton starting from initial capacitor voltage V (0), the terminal voltage V (t) follows: V (t) = V∞ + (V (0) − V∞) e−t/ , where V∞ = Voc − Iload Rint (23) Evaluating Equation (23) across four candidate drive regimes reveals the exact physical con- straints (Table 7):

1. 32 mA for 100 ms (23.8): Decays completely to steady-state V∞ = 2.95 − (0.032 × 42) = 1.606 V, falling below the 1.70 V nRF52811 brownout threshold!

2. Isolated 2.0 ms Micro-Chirp at 32 mA: Starting from V (0) = 2.95 V, voltage at t = 2.0 ms reaches 1.606 + (2.95 − 1.606)e−2.0/4.2 = 2.440 V (not 2.68 V). Under repeated 2 ms ON / 8 ms OFF periodic pulsing, the steady-state periodic minimum settles at 2.38 V.

3. Current-Limited 12.0 mA Alert Load: Limiting the combined piezo/LED boost stage to Iload = 12.0 mA yields a steady-state voltage of V∞ = 2.95 − (0.012 × 42) = 2.446 V > 1.70 V even for continuous or 100 ms pulses.

Table 6: Wearable Current Consumption and Analytical CR2450 Runtime Estimates

Operating mode	Interval / duty	Peak	Mean	Basis / qualification
RTC sleep + 32.768 kHz crystal + RAM	Continuous	1.80 μA	1.80 μA	nRF52811 system-on sleep specification
BLE TX, Config A (LE 1M, +4 dBm)	1.15 ms / 800 ms	7.00 mA	10.06 μA	nRF52811; includes RF airtime and ramp-up
BLE TX, Config B (coded S=8, +8 dBm)	5.80 ms / 800 ms	14.20 mA	102.95 μA	nRF52833 extended advertising
Baseline sparse wake scan	5 ms / 4,000 ms	4.60 mA	5.75 μA	Insufficient overlap with 50 ms wake
Deterministic wake scan	22.5 ms / 4,000 ms	4.60 mA	25.88 μA	Designed to overlap 20 ms wake advertisement
AES-CMAC rotation	Every 900 s	3.80 mA	< 0.01 μA	Hardware AES-128 accelerator
Baseline total	Continuous	—	29.74 μA	Idealized 28.5 months at 620 mAh; 20.0 months at 434 mAh
Reconciled Config A total	Continuous	—	37.74 μA	nRF52811 +4 dBm and deterministic scan
Reconciled Config B total	Continuous	—	130.63 μA	nRF52833 +8 dBm coded PHY and scan
Rack storage deep sleep	Between trips	0.60 μA	0.60 μA	Reed switch disables sensor / scan rails

4. BLE Radio TX Spike (7.0 mA for 1.15 ms): For the 1.15 ms nRF52811 +4 dBm TX event (V∞ = 2.656 V), the 100 µF capacitor holds V (1.15 ms) = 2.880 V (V = 70 mV).

Table 7: Thevenin RC Transient Voltage Audit at −5 ◦ C (Voc = 2.95 V, Rint = 42 Ω, C = 100 µF, = 4.20 ms, Claim C08) Load Regime Load Current Pulse (ton) Steady V∞ Min V (ton) Brownout Status (> 1.70 V Cutoff) 1. Baseline 100 ms Alert Pulse 32.0 mA 100.0 ms 1.606 V 1.606 V FAILS (< 1.70 V; refutes 2.68 V claim) 2. Isolated 2 ms Micro-Chirp 32.0 mA 2.0 ms 1.606 V 2.440 V Analytical pass (> 1.70 V); needs bench test 3. Periodic 2 ms/10 ms Chirp 32.0 mA 2.0 ms 1.606 V 2.380 V Analytical pass; requires cold rail trace 4. Current-Limited Alert Stage 12.0 mA 100.0 ms 2.446 V 2.446 V Safe in DC steady state (V = 0.504 V) 5. BLE TX Event (Config A) 7.0 mA 1.15 ms 2.656 V 2.880 V Buffered by 100 µF cap (V = 70 mV)

7 Ethical, Legal (DPDP Act 2023), and Physical Safety Safeguards

Because Rakshasutra processes localization events for children and cognitively vulnerable older adults, the system enforces three mandatory safeguards:

1. Statutory Compliance with India's DPDP Act, 2023 (Section 9): Processing personal data of children and protected adults requires verifiable guardian consent and prohibits behav- ioral profiling [8]. Explicit guardian consent is recorded at counter pairing, zero personally identifiable information (PII) is broadcast over BLE or LoRaWAN, and all historical pole check-in logs are automatically purged within 24 to 72 hours of exit unpairing.

2. Phone-Free Sanctum Kiosk Access Controls: Touch kiosks inside phone-restricted zones (Bhawan and Adhkuwari) require physical tap authentication using the family's own RFID Yatra Parchi or Caregiver Tag. Kiosks prohibit text searches by name, lock out after 3 invalid scans, blank their privacy-filtered screens after 30 s, and explicitly display the age of the last pole observation so staff and caregivers never mistake a stale zone report for live GPS tracking (Claim C18).

3. Button-Battery Ingestion Protection (IEC 60086-4): Coin cells pose severe chemical burn hazards if swallowed [18]. Wearables utilize an ultrasonically welded (or captive Torx tool- only) IP67 enclosure with no user-accessible battery door, paired with medical-grade TPU straps sanitized in a UV-C chamber between trips.

8 Summary of Audit Ledger and Future Validation Roadmap

Table 8 summarizes the complete Day 1 science audit ledger across all core claims ('C01–C18'), establishing a transparent baseline for ongoing discrete-event simulation and hardware bench testing.

Table 8: Condensed Audit Reconciliation Ledger (Claims C01–C18) ID Topic Baseline v5.0 Claim Reconciled v5.1 Analytical Status C01 Route Poles 255 poles (Zone 4: 6.5 km = 25 poles) 255 partial (hubs) vs. 295 full (65 poles in Zone 4 at 100 m) C02/03 BLE PHY & TX +8 dBm nRF52811; Legacy Coded PHY Split into Config A (nRF52811 +4 dBm LE 1M Legacy) & Config B (nRF52833 +8 dBm Coded S=8 Extended) C04/05 Scan Overlap 5 ms async scan (2.48% hit over 4 scans) Crystal-synced 14 ms tether window + 22.5 ms scan vs. 20 ms pole wake (100% geometric overlap) C06/07 Power & Life 29.74 µA; 11.1 mos at −5 ◦ C 37.74 µA (Config A); explicit capacity ratios (22.5 mos nominal; 8.7–15.7 mos cold derated) C08 −5 ◦ C RC Pulse 100 µF holds 32 mA/100 ms at 2.68 V Refuted (= 4.2 ms → 1.606 V); solved via 12 mA current limit (2.446 V) or 2 ms chirps (2.38–2.44 V) C09/10 LoRa Surge 15.8% loss (G = 8.6%) at 60 poles; fixed DR-adaptive chunks; 60-pole loss is 200B 28.99% (full), 21.35% (implicit dep.), and 14.77% (40-pole fiber offload) C11/12 Bloom & Crypto 64B Bloom (2.6% FP); cleartext ci Set-intersection rule (Sdwell Sauth); 16-bit ephemeral nonce rk replaces cleartext ci C13–18 Evidence Labels Described RF/EV/battery as "measured" Strictly classified as analytical estimates under stated assumptions pending bench/field trials

2 Conclusion

We presented the architectural design, mathematical modeling, and rigorous audit reconcilia- tion of Rakshasutra (SurakshaNet), a Zero-UI cognitive-assistive wearable and edge-filtered BLE–LoRaWAN localization network for older adults with memory decline and young children in cellular-denied mountain environments. By reconciling BLE Link Layer PHY constraints, deterministic scan-window overlap conditions, Thevenin RC cold-cell transient bounds, and DR-adaptive LoRaWAN IN865 congestion control, this work establishes a transparent, math- ematically defensible foundation for assistive crowd-safety systems.

References

[1] LoRa Alliance Technical Committee, RP002-1.0.5 LoRaWAN Regional Parameters (IN865–867 MHz Specification), Fremont, CA, USA: LoRa Alliance, 2024.

[2] Semtech Corporation, LoRa Modem Designer's Guide (AN1200.13) & LoRaWAN Time-on-Air Calculation (AN1200.22), Camarillo, CA, USA, 2020.

[3] J. Song, R. Poovendran, J. Lee, and T. Iwata, "The AES-CMAC Algorithm," Internet Engineer- ing Task Force (IETF) RFC 4493, Jun. 2006.

[4] Bluetooth Special Interest Group (SIG), Bluetooth Core Specification Version 5.1 (LE Coded PHY and Advertising Extensions), Kirkland, WA, USA, 2019.

[5] Bluetooth Special Interest Group (SIG), Bluetooth Core Specification Version 6.0 / 6.1, Vol. 6: Low Energy Controller (Link Layer Specification), Kirkland, WA, USA, 2024.

[6] B. H. Bloom, "Space/Time Trade-offs in Hash Coding with Allowable Errors," Commun. ACM, vol. 13, no. 7, pp. 422–426, Jul. 1970.

[7] A. Kirsch and M. Mitzenmacher, "Less Hashing, Same Performance: Building a Better Bloom Filter," Random Struct. Algorithms, vol. 33, no. 2, pp. 187–218, Sep. 2008.

[8] Ministry of Law and Justice, Government of India, The Digital Personal Data Protection Act, 2023 (No. 22 of 2023), Section 9: Processing of Personal Data of Children, New Delhi, India, Aug. 2023.

[9] Wireless Planning and Coordination (WPC) Wing, Ministry of Communications, Govern- ment of India, Use of Low Power Equipment in the Frequency Band 865–867 MHz for RFID and IoT Applications (G.S.R. 564(E)), New Delhi, India.

[10] A. Augustin, J. Yi, T. Clausen, and W. M. Townsley, "A Study of LoRa: Long Range & Low Power Networks for the Internet of Things," Sensors, vol. 16, no. 9, p. 1466, Sep. 2016.

[11] M. Centenaro, L. Vangelista, A. Zanella, and M. Zorzi, "Long-Range Communications in Unlicensed Bands: The Rising Stars in the IoT and Smart City Scenarios," IEEE Wireless Commun., vol. 23, no. 5, pp. 60–67, Oct. 2016.

[12] F. Adelantado, X. Vilajosana, P. Tuset-Peiro, B. Martinez, J. Melia-Segui, and T. Watteyne, "Understanding the Limits of LoRaWAN," IEEE Commun. Mag., vol. 55, no. 9, pp. 34–40, Sep. 2017.

[13] M. C. Bor, U. Roedig, T. Voigt, and J. M. Alonso, "Do LoRa Low-Power Wide-Area Networks Scale?" in Proc. ACM MSWiM, Nov. 2016, pp. 59–67.

[14] A. Heinrich, M. Stute, T. Kornhuber, and M. Hollick, "Who Can Find My Devices? Security and Privacy of Apple's Crowd-Sourced Bluetooth Location Tracking System," Proc. Privacy Enhancing Technol. (PoPETs), vol. 2021, no. 3, pp. 227–245, Jul. 2021.

[15] Nordic Semiconductor ASA, nRF52811 Product Specification v1.1 (Key Features and Ra- dio Electrical Specification) and nRF52833 Product Specification, Trondheim, Norway, 2020–2024.

[16] Panasonic Energy Co., Ltd., Lithium Manganese Dioxide Coin Battery CR2450 Datasheet (620 mAh Nominal Specification), Osaka, Japan, 2024.

[17] Semtech Corporation, SX1261/2 Long Range, Low Power, Sub-GHz RF Transceiver Datasheet Rev. 2.1, Camarillo, CA, USA, 2021.

[18] International Electrotechnical Commission (IEC), IEC 60086-4: Primary Batteries—Part 4: Safety of Lithium Batteries (Child-Resistant Packaging and Button Cell Ingestion Safeguards), Geneva, Switzerland, 2019.